Underwater Image Restoration and Marine Life Detection

Milestone 1 Submission

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1. Project Title

Underwater Image Restoration and Marine Life Detection: An Investigation into Detection-Driven Enhancement

2. Group Details

Role	Name	University Registration No.
Group Leader	Gyanendra Prakash	S24CSEU0771
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Table 1. Group members and roles.

3. Problem Domain Identification and Analysis

3.1 Problem Definition

Underwater images captured for marine research and monitoring suffer from severe, non-uniform degradation caused by the way water absorbs and scatters light. Longer wavelengths (red) are attenuated first, followed by green, producing a characteristic blue-green colour cast, haze, and low contrast that worsen with depth and distance from the camera. Because this distortion depends on turbidity, depth, and ambient lighting, it cannot be corrected with a single fixed transformation – each image effectively requires its own adaptive correction. This degradation directly harms the accuracy of any automated vision task built on such footage, including marine biology surveys, coral reef health monitoring, autonomous underwater vehicle (AUV) navigation, and fisheries research, all of which depend on reliably detecting fish, coral, or other marine life in raw video and image data.

3.2 Significance and Objectives

- Design and implement a restoration pipeline that corrects underwater colour distortion and hazeacross varying water conditions.
- Use the restored images to improve the accuracy of automatic marine species and object detectionmodels.
- Investigate, as the project's central research question, whether restoration optimised purely for visualquality actually improves object detection accuracy – several recent papers indicate this is not guaranteed – making the project a genuine research comparison rather than a single enhancement pipeline.

4. Review of Existing Systems

4.1 Related Datasets

UIEB (Underwater Image Enhancement Benchmark) – introduced by Li et al. (IEEE Transactions on Image Processing, 2019) to address the lack of real-world evaluation data in prior underwater enhancement research. It contains 950 real-world underwater images, 890 of which have paired high-quality reference images, plus 60 additional challenging images without references, spanning coral reefs, marine fauna, and shipwreck scenes. The same paper introduces Water-Net, a CNN trained on UIEB that serves as a standard baseline model in this field.

EUVP (Enhancement of Underwater Visual Perception dataset) – introduced by Islam, Xia, and Sattar (IEEE Robotics and Automation Letters, 2020). It provides over 12,000 paired and 8,000 unpaired images captured with seven different cameras during real ocean exploration and human-robot interaction experiments, offering broader variability in camera hardware and conditions than UIEB.

4.2 Related Methods

- **Water-Net** (Li et al., 2019) – a CNN-based enhancement model trained directly on UIEB; the standard academic baseline for underwater enhancement.
- **FUnIE-GAN** (Islam, Xia, and Sattar, 2019/2020) – a conditional GAN designed for fast, real-time underwater image enhancement, trained with a combined loss covering content, colour, texture, and style. Its authors show that enhanced images improve downstream performance on object detection, pose estimation, and saliency prediction, supporting the idea that restoration quality and detection performance are linked but not automatically aligned.
- **CycleGAN-based approaches** – enable unpaired underwater enhancement, removing the need for perfectly matched clean/degraded image pairs, which are difficult to obtain for real underwater scenes.
- **DUnIE-GAN** ("A Generative Approach for Detection-driven Underwater Image Enhancement") – compares perceptually-focused enhancement (such as FUnIE-GAN) against detection-driven enhancement, and shows that higher perceptual-quality scores do not necessarily produce more accurate detection results, providing direct evidence for the research gap this project targets.
- A recent **dual-branch framework** for joint optimisation of underwater image enhancement and object detection frames enhancement and detection as a single joint optimisation problem rather than two independent stages, using UIEB, EUVP, URPC2022, and DUO datasets – confirming this remains an open, active research direction.

4.3 Limitations of Existing Work and Identified Research Gap

- Most classical and early deep-learning enhancement methods (CLAHE, Retinex-based methods, basicCNNs) optimise for human perceptual quality using metrics such as PSNR and SSIM against a reference image, but multiple recent studies show this does not guarantee improved object detection performance downstream.
- Enhancement and detection are typically trained and evaluated as two separate, disconnected stagesrather than jointly; joint optimisation is only beginning to be explored in 2025-2026 literature.

- Most published work evaluates on a single dataset (usually UIEB or EUVP alone), so generalisationacross different water types, turbidity levels, and camera setups remains under-studied.
- Real-time or lightweight deployment suitable for actual underwater robots and AUVs is addressed byonly a small number of methods, with FUnIE-GAN a notable exception explicitly designed for real-time use.

Project angle: rather than building a restoration model and separately running a detector on its output, this project will explicitly measure and compare whether restoration optimised for perceptual quality (PSNR/SSIM) actually improves detection accuracy (mAP), and whether a detection-driven training objective can do better. This reframes the project from "building a pipeline" into a measurable research question.

5. Project Feasibility

5.1 Proposed Deep Learning Model

Stage 1 – Restoration

- Classical baseline: CLAHE, per-channel histogram equalisation, and simple dehazing via theatmospheric scattering model, giving a cheap, quick comparison floor.
- Deep model: fine-tune a Water-Net or FUnIE-GAN style architecture on UIEB and/or EUVP.FUnIE-GAN is lighter and faster to train; Water-Net is the more established academic baseline. One will be selected as the primary model, with the other used for comparison.

Stage 2 – Detection

A standard object detector (YOLOv8 or similar) will be fine-tuned for marine species and object detection using a labelled underwater detection dataset such as DUO, Brackish, or a suitable fish-specific dataset (final choice to be confirmed).

Stage 3 – Comparative Evaluation (core research contribution)

- Run detection on raw degraded images, classically-enhanced images, and deep-model enhancedimages.
- Report detection accuracy (mAP) at each stage rather than relying only on visual before/aftercomparison.
- If time permits, introduce a detection-aware loss term during restoration training (following theDUnIE-GAN approach) and compare it against perceptual-quality-only restoration.

5.2 Dataset to be Used

Dataset	Purpose	Source
UIEB	Restoration training/evaluation (paired real images)	li-chongyi.github.io/proj_benchmark.html
EUVP	Restoration training/evaluation (larger paired + unpaired set)	irvlab.cs.umn.edu/resources/euvp-dataset

Underwater object detection dataset (DUO / Brackish / similar)	Detection training/evaluation	To be finalised; licensing to be confirmed

Table 2. Datasets to be used for restoration and detection.

5.3 Implementation Plan (Approx. 8–10 Weeks)

Week(s)	Activity
1-2	Finalise scope, download UIEB and EUVP, source a labelled detection dataset, set up GPU environment (Colab/Kaggle).
3	Implement classical restoration baseline (CLAHE / dehazing) for an initial comparison point.
4-5	Train the deep restoration model (FUnIE-GAN or Water-Net) on UIEB/EUVP.
6	Fine-tune a YOLO-style detector on the chosen marine detection dataset.
7	Run the core comparison: detection accuracy on raw vs. classically-enhanced vs. deep-enhanced images
8	(If time allows) Implement and evaluate a detection-aware restoration loss.
9-10	Buffer for write-up, visualisations (before/after grids, mAP comparison charts), and evaluation preparation.

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6. References

References 1–7 are the original Milestone 1 references. References 8–17 are additional recent (2023–2026) references added in this update; they were sourced and verified against arXiv, DOI, and publisher listings, and directly support the detection-driven enhancement research angle described in Sections 3.2 and 4.3.

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